

INDIVIDUAL ASSIGNMENT 1

ML System Design Document

Course	DDM501 - AI in DevOps, DataOps, MLOps
Assignment	Individual Assignment 1
Weight	5%
Submission	Written report

I. Overview

This assignment requires to analyze a real-world problem that could benefit from a Machine Learning solution and create an ML system design document. The concepts are applied from *Sessions 1-2 covering requirements engineering, goal setting, metrics selection, and system architecture design.*

The purpose is to demonstrate your ability to think critically about ML system development from a production perspective, considering not just the model but the entire system lifecycle, stakeholders, constraints, and trade-offs.

II. Guidelines

You are free to choose your ML application scenario. However, your chosen scenario **must** meet the following criteria:

2.1. Required Characteristics

- ML Problem:** The problem must require machine learning tasks such as predictions, classifications, recommendations, anomaly detection, etc..
- Stakeholders:** Assume that your scenario must involve at least 2 distinct stakeholders with different goals and concerns such as end users, business owners, operations team, data scientists.
- Context:** The system should be designed for real-world deployment, not just research or experimentation and consider some aspects such as scalability, reliability, and maintenance.
- Datasets:** Describe realistic data sources, or might be hypothetical. Consider data collection, quality, and privacy concerns.
- Measurable Business Impact:** The ML system should provide some business values that can be measured.

2.2. Complexity Level

The scenario should be:

- Not too simple:** Avoid basic single-model applications with obvious solutions (e.g., simple spam classifier with no production considerations)
- Not too complex:** Avoid systems requiring multiple interacting ML models or extremely specialized domain knowledge

2.3. Examples

Below are examples to help you to select:

Category	Example Scenarios
E-commerce & Retail	Product recommendation system, demand forecasting, price optimization, customer churn prediction, fraud detection

Healthcare	Patient risk prediction, medical image analysis, drug interaction detection
Finance	Credit scoring, transaction fraud detection, algorithmic trading signals, loan default prediction
Manufacturing	Predictive maintenance, quality control, supply chain optimization, defect detection
Transportation	Route optimization, demand prediction, driver behavior analysis, ETA estimation
Content & Media	Content recommendation, sentiment analysis, content moderation, personalized search
HR & Workforce	Resume screening, employee attrition prediction, skill matching, workforce scheduling
IoT & Smart Systems	Energy consumption prediction, anomaly detection in sensor data, smart home automation

III. Report Requirements

The report includes the following sections.

Section	Requirements
Problem definition (20%)	Clearly define the problem you are addressing: - Context and Background: Describe the domain, organization context, and why this problem matters - Problem Statement: State the problems clearly and in measurable terms - Current situation: How is this problem currently being handled - Justification: Why is ML the appropriate approach - Stakeholder Identification: List all stakeholders and their concerns
Requirements analysis (20%)	Requirements for your ML system: <i>Functional Requirements</i> - Core ML functionality (what predictions/decisions the system makes) - Input/Output specifications - Integration requirements with existing systems - User interaction requirements <i>Non-Functional Requirements</i> - Performance: Latency, throughput, response time expectations - Scalability: Expected load, scaling strategy - Reliability: Uptime requirements, failure handling, graceful degradation

	- Maintainability: Update frequency, model retraining needs, monitoring requirements <i>Data Requirements</i> - Data sources - Data quality - Privacy - Data volume
Goals and metrics (20%)	Define a clear hierarchy of goals and corresponding metrics: <i>Business goals</i> <i>System goals</i> <i>Model goals</i> <i>Acceptable thresholds and baselines</i>
High-level architecture design (25%)	<i>Present a high-level system architecture:</i> clear architecture diagrams data flow through the system ML pipeline stages (data ingestion, preprocessing, training, inference) <i>Component Description:</i> Purpose and responsibility Technologies/tools considered
Trade-offs analysis (15%)	Analyze at least 3-4 significant trade-offs in design: <i>Accuracy vs. Latency:</i> Complex models may be more accurate but slower <i>Freshness vs. Cost:</i> Real-time updates are expensive but provide current predictions <i>Simplicity vs. Performance:</i> Simpler systems are easier to maintain but may underperform <i>Automation vs. Control:</i> Fully automated systems may make unexpected decisions <i>Privacy vs. Personalization:</i> More data enables better predictions but raises privacy concerns

IV. Submission Guidelines

4.1. Submission Process

- **File naming:** DDM501_Assignment1_[StudentID]_[Name].pdf